

## Day 1(LOS)

### Part A — User and System Investigation (10 minutes)

1.Determine the username of the account you are currently using.

```
msis@master:~/Day1$ whoami
\msis
```

2.Display your UID, primary GID, and supplementary groups.

```
msis@master:~/Day1$ id
uid=1000(msis) gid=1000(msis) groups=1000(msis),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users),114(lpadmin),126(libvirt)
```

3.Display all users currently logged in to the system.

```
msis@master:~/Day1$ who
msis      seat0      2026-09-01 09:18 (login screen)
msis      tty2        2026-09-01 09:18 (tty2)
```

4.Determine what activities the currently logged-in users are performing.

```
msis@master:~/Day1$ w
09:43:08 up 26 min, 1 user, load average: 0.27, 0.38, 0.35
USER      TTY      FROM          LOGIN@      IDLE        JCPU        PCPU        WHAT
msis      tty2      -             09:18       26:10       0.01s       0.01s       /usr/li
```

5.Examine /etc/passwd and identify the following information for your current account:

Username

UID

GID

Home directory

Login shell

```
msis@master:~/Day1$ grep "^$(whoami):" /etc/passwd
msis:x:1000:1000:MSIS:/home/msis:/bin/bash
```

6.Determine how many user-account entries are present in /etc/passwd.

```
msis@master:~/Day1$ wc -l /etc/passwd
58 /etc/passwd
```

7.Display only the username and UID fields from /etc/passwd.

```
msis@master:~/Day1$ cut -d: -f1,3 /etc/passwd
root:0
daemon:1
bin:2
sys:3
sync:4
games:5
man:6
lp:7
mail:8
news:9
uucp:10
proxy:13
www-data:33
backup:34
list:38
irc:39
_apt:42
nobody:65534
systemd-network:998
systemd-timesync:996
dhcpcd:100
messagebus:101
syslog:102
systemd-resolve:991
uidd:103
usbmux:104
tss:105
systemd-oom:990
kernoops:106
whoopsie:107
dnsmasq:999
avahi:108
tcpdump:109
sssd:110
speech-dispatcher:111
cups-pk-helper:112
fwupd-refresh:989
saned:113
geoclue:114
cups-browsed:115
hplip:116
gnome-remote-desktop:988
polkitd:987
rtkit:117
```

**8. Find all accounts whose UID is 0. Explain the security significance of UID 0.**

**Compare /etc/passwd and /etc/shadow. Why are password hashes stored in /etc/shadow instead of /etc/passwd?**

```
msis@master:~/Day1$ awk -F: '$3 == 0 {print $1}' /etc/passwd
root
```

Answer and explanation:

UID 0 represents the superuser (root). Any account with UID 0 has full administrative privileges.

Therefore, unexpected accounts with UID 0 are a major security risk because they can control the entire system.

```
msis@master:~/Day1$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
```

```
msis@master:~/Day1$ sudo cat /etc/shadow
[sudo] password for msis:
root:*:20305:0:99999:7:::
daemon:*:20305:0:99999:7:::
bin:*:20305:0:99999:7:::
sys:*:20305:0:99999:7:::
sync:*:20305:0:99999:7:::
games:*:20305:0:99999:7:::
man:*:20305:0:99999:7:::
lp:*:20305:0:99999:7:::
mail:*:20305:0:99999:7:::
news:*:20305:0:99999:7:::
uucp:*:20305:0:99999:7:::
proxy:*:20305:0:99999:7:::
www-data:*:20305:0:99999:7:::
backup:*:20305:0:99999:7:::
list:*:20305:0:99999:7:::
irc:*:20305:0:99999:7:::
_apt:*:20305:0:99999:7:::
nobody:*:20305:0:99999:7:::
systemd-network:!*:20305:0:0:0:
systemd-timesync:!*:20305:0:0:0:
```

Answer:

/etc/passwd:

Contains user account information.

Contains username, UID, GID, home directory, and login shell.

Usually readable by all users.

/etc/shadow:

Contains password hashes.

Contains password expiration and aging information.

Accessible only to root or authorized users.

Why are password hashes stored in /etc/shadow?

Password hashes are stored in /etc/shadow for security. If hashes were publicly readable, attackers could copy them and attempt to crack passwords offline.

## Part B — Creating and Verifying Users (20 Minutes)

1. Create the following users and ensure that a home directory is created for each:

alice, bob, charlie

```
sanjana@sanjana-VMware-Virtual-Platform:~/Day1$ sudo useradd -m alice
[sudo: authenticate] Password:
sanjana@sanjana-VMware-Virtual-Platform:~/Day1$ sudo useradd -m bob
sanjana@sanjana-VMware-Virtual-Platform:~/Day1$ sudo useradd -m charlie
```

The -m option creates a home directory.

2. Verify that all three accounts have been added to /etc/passwd.

```
sanjana@sanjana-VMware-Virtual-Platform:~/Day1$ grep -E '^(alice|bob|charlie):'
/etc/passwd
alice:x:1001:1002::/home/alice:/bin/sh
bob:x:1002:1003::/home/bob:/bin/sh
charlie:x:1003:1004::/home/charlie:/bin/sh
```

### 3. Determine the UID assigned to each newly created user.

```
sanjana@sanjana-VMware-Virtual-Platform:~/Day1$ id -u alice
1001
sanjana@sanjana-VMware-Virtual-Platform:~/Day1$ id -u bob
1002
sanjana@sanjana-VMware-Virtual-Platform:~/Day1$ id -u charlie
1003
```

### 4. Determine the home directory and default shell assigned to alice.

Find where Alice's personal files are stored and which shell she uses when logging in.

```
sanjana@sanjana-VMware-Virtual-Platform:~/Day1$ getent passwd alice
alice:x:1001:1002::/home/alice:/bin/sh
```

Answer:

Home directory: /home/alice

Default shell: /bin/bash

### 5. Inspect /home and verify that the required home directories were created.

To check if directories were created for all three users.

```
sanjana@sanjana-VMware-Virtual-Platform:~/Day1$ ls -l /home
total 16
drwxr-x--- 2 alice  alice  4096 Sep  1 16:53 alice
drwxr-x--- 2 bob    bob    4096 Sep  1 16:53 bob
drwxr-x--- 2 charlie charlie 4096 Sep  1 16:57 charlie
drwxr-x--- 16 sanjana sanjana 4096 Sep  1 16:49 sanjana
```

### 6. Set passwords for all three accounts.

Assign login passwords to the newly created users.

```
sanjana@sanjana-VMware-Virtual-Platform:~/Day1$ sudo passwd alice
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
```

### 7. Switch from your current account to alice using a login shell.

Log in as Alice while loading her normal login environment.

```
sanjana@sanjana-VMware-Virtual-Platform:~$ su - alice
Password:
$ whoami
alice
$
```

Explanation:

The - creates a login shell and loads Alice's login environment, including her home directory and environment settings.

### 8. Verify the change using an appropriate command, and then return to your original account.

```
$ id
uid=1001(alice) gid=1002(alice) groups=1002(alice)
$ exit
```

## Part C — Group Management and Access Control (30 Minutes)

### 1. Create two groups: security and developers

Create two groups that can be used to organise users and control access

```
msis@master:~/Day1$ sudo groupadd security
msis@master:~/Day1$ sudo groupadd developers
```

### 2. Add alice and charlie to the security group.

Make two users supplementary members of the security group

```
msis@master:~/Day1$ sudo usermod -aG security alice
msis@master:~/Day1$ sudo usermod -aG security charlie
```

Usermod modifies an existing user

-G specifies supplementary groups

-a appends the group and preserves existing group memberships

3.Add bob to the developers group.

```
msis@master:~/Day1$ sudo usermod -aG developers bob
```

4.Without switching accounts, verify the group membership of all three users.

```
msis@master:~/Day1$ id alice
uid=1004(alice) gid=1004(alice) groups=1004(alice),1007(security)
msis@master:~/Day1$ id bob
uid=1005(bob) gid=1005(bob) groups=1005(bob),1008(developers)
msis@master:~/Day1$ id charlie
uid=1006(charlie) gid=1006(charlie) groups=1006(charlie),1007(security)
```

5.Display the entries corresponding to security and developers from /etc/group.

```
msis@master:~/Day1$ grep -E '^(security|developers):' /etc/group
security:x:1007:alice,charlie
developers:x:1008:bob
```

6.Add bob to the security group without removing him from the developers group.

```
msis@master:~/Day1$ sudo usermod -aG security bob
```

7.Verify that bob now belongs to both groups.

```
msis@master:~/Day1$ groups bob
bob : bob security developers
```